Self-scaled bound of the regular cuboctahedron

We provide an exact certificate for the self-scaled bound of the slack matrix X of the regular cuboctahedron. We use the matrix `Scuboctahedron` in `SlackMat.mat`, with the row and column ordering of the accompanying repository. This 12×14 matrix has 48 zeros, 72 entries equal to 1 and 48 entries equal to 2. Its first six columns correspond to square facets, and its last eight columns to triangular facets. Among the entries equal to 2, there are 24 in each class of columns.

Recall that

$$\mathrm{SSB}(X) = \max_{L}\{\langle L, X\rangle : \langle L, R\rangle \le 1 \text{ for all } R \in \mathcal{R}(X)\}, \qquad \mathcal{R}(X) = \{uv^\top : u, v \ge 0,\ uv^\top \le X\}.$$

The script `CertificateCuboctahedron.py` checks the exact geometry, symmetries, finite enumeration and decomposition below using integer arithmetic.

Proposition 1. *The slack matrix X satisfies* $\mathrm{SSB}(X) = 39/5$. *An optimal dual certificate is*

$$L^*_{ij} = \begin{cases} 0 & \text{if } X_{ij} = 0, \\ 3/40 & \text{if } X_{ij} = 1, \\ 1/24 & \text{if } X_{ij} = 2 \text{ and column } j \text{ is a square facet}, \\ 1/120 & \text{if } X_{ij} = 2 \text{ and column } j \text{ is a triangular facet}. \end{cases}$$

Proof. The objective value of L^* is

$$\langle L^*, X\rangle = 72\frac{3}{40} + 24\frac{2}{24} + 24\frac{2}{120} = \frac{27}{5} + 2 + \frac{2}{5} = \frac{39}{5}.$$

We first verify its feasibility. For a nonzero admissible matrix, normalize its factors so that $\max_i u_i = 1$. Then $0 \le u_i \le 1$ and $0 \le v_j \le 2$. Since $L^* \ge 0$, we may increase every positive v_j to at least 1, and then every positive u_i to at least $1/2$, without decreasing the objective. These changes preserve admissibility: on the positive supports, every entry of X is at least 1, while a changed product is at most 1.

Write $u_i = 2^{-a_i}$ and $v_j = 2^{b_j}$ on the positive supports, with $a_i, b_j \in [0, 1]$. The constraints at $X_{ij} = 2$ are automatic; those at $X_{ij} = 1$ become $b_j \le a_i$. For $\theta \in (0, 1)$, define

$$a_i(\theta) = \mathbf{1}_{\{a_i \ge \theta\}}, \qquad b_j(\theta) = \mathbf{1}_{\{b_j \ge \theta\}}.$$

These binary vectors preserve the inequalities and have average (a, b). Since

$$f(a, b) = \sum_{i \in \mathrm{supp}(u),\, j \in \mathrm{supp}(v)} L^*_{ij} 2^{b_j - a_i}$$

is convex, Jensen's inequality gives $f(a, b) \le \int_0^1 f(a(\theta), b(\theta))\, d\theta$. An index with $u_i = 1$ remains fixed. Thus it suffices to consider

$$u_i \in \{0, 1/2, 1\}, \qquad v_j \in \{0, 1, 2\}.$$

The 48 signed coordinate permutations of the cuboctahedron induce a group G of simultaneous row and column permutations preserving X. They preserve facet types, hence also L^*, and act transitively on the rows. Therefore we may assume $u_1 = 1$. The first row contains four zeros, six ones and four twos. Accordingly, v_j is zero at the zeros, belongs to $\{0, 1\}$ at the ones, and to $\{0, 1, 2\}$ at the twos. There are

$$2^6 3^4 = 5\,184$$

configurations. For each of them, choose every u_i as its largest feasible value in $[0, 1]$: it is zero if some positive v_j meets a zero of row i, is $1/2$ if no such zero occurs but some $v_j = 2$ meets $X_{ij} = 1$, and is 1 otherwise. Put $C = 120L^*$ and $\widehat{u} = 2u$. Then

$$240\langle L^*, uv^\top \rangle = \widehat{u}^\top C v.$$

All entries in this expression are integers. The exhaustive enumeration gives a maximum of exactly 240, proving feasibility and hence $\mathrm{SSB}(X) \geq 39/5$.

For the upper bound, define three atoms $R_A = u_A v_A^\top$, $R_B = u_B v_B^\top$ and $R_C = u_C v_C^\top$ by

$$\begin{array}{lll}
(u_A)_i = 1 & & i \in \{1, 2, 3, 6\}, \\
(v_A)_6 = 2, & (v_A)_j = 1 & j \in \{7, 10, 11, 14\}, \\
(u_B)_1 = 1, & (u_B)_i = 1/2 & i \in \{2, 3, 4, 5\}, \\
(v_B)_j = 2 & & j \in \{3, 6, 7, 11\}, \\
(u_C)_i = 1 & & i \in \{1, 3, 9, 12\}, \\
(v_C)_j = 1 & & j \in \{1, 10, 13, 14\},
\end{array}$$

with all other entries zero. Indices start at 1. The script verifies $0 \leq R_A, R_B, R_C \leq X$ and the exact identity

$$X = \sum_{g \in G} \left(\frac{R_A^g}{80} + \frac{R_B^g}{10} + \frac{R_C^g}{20} \right).$$

Consequently, every feasible L satisfies

$$\langle L, X \rangle \leq 48 \left(\frac{1}{80} + \frac{1}{10} + \frac{1}{20} \right) = \frac{39}{5}.$$

Together with the lower bound, this proves the equality. □

Verification Run `python CertificateCuboctahedron.py`. Only NumPy is required. The script is self-contained: it constructs the exact slack matrix from standard coordinates of the cuboctahedron, enumerates its 48 signed coordinate symmetries, and verifies both bounds without an optimization solver or an external data file. Run Python without the `-O` option, since the verification uses assertions.